Information Retrieval with BM25 and Embeddings

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0. Imports and helper functions

In [1]:
```
import gzip
import json
import pickle
import re
import string
import pandas as pd
from IPython.display import display, Markdown
from textwrap import shorten

# Get the root
import sys
import os
project_root = os.path.abspath(os.path.join(os.getcwd(), ".."))
sys.path.append(project_root)

# Suppress warnings
import warnings
warnings.filterwarnings('ignore')

#import nltk
#nltk.download("stopwords")
from nltk.corpus import stopwords
from rank_bm25 import BM25Okapi
from langchain_core.documents import Document
from sentence_transformers import SentenceTransformer

# Import scripts
from src.bm25 import build_bm25, save_bm25, load_bm25, bm25_search
from src.semantic import (
    build_embeddings,
    build_faiss_index,
    save_faiss,
    load_faiss,
    semantic_search
)
```

In [2]:
```
def load_jsonl_gz(path, n=200):
    '''
    load first N records from a .jsonl.gz file
    '''
    records = []
    with gzip.open(path, "rt", encoding="utf-8") as f:
```

```
        for i, line in enumerate(f):
            if i >= n:
                break
            try:
                records.append(json.loads(line))
            except json.JSONDecodeError:
                continue
    return records
```

In [3]:

```
def display_search_results(results, query, system_name, max_chars=200):
    '''
    Pretty display for retrieval results.
    '''
    rows = []

    for rank, r in enumerate(results, start=1):
        metadata = r["metadata"]
        source = metadata["source"]

        raw_text = r["text"].strip()
        parts = [p.strip() for p in raw_text.split("\n\n") if p.strip()]

        title = parts[0]
        body = " ".join(parts[1:]) if len(parts) > 1 else raw_text
        body = " ".join(body.split())

        if source == "metadata":
            categories = metadata.get("categories", [])
            categories_str = " > ".join(categories[:4]) if isinstance(
                categories, list) else ""
            info = metadata.get("main_category", "")
            if categories_str:
                info = f"{info} | {categories_str}" if info else categories_
        else:
            info = f'rating={metadata.get("rating", "")}, verified={metadata

        rows.append({
            "Rank": rank,
            "Score": float(r["score"]),
            "Source": source,
            "ASIN": metadata["asin"],
            "Title": title,
            "Info": info,
            "Snippet": shorten(body, width=max_chars, placeholder="...")
        })

    df = pd.DataFrame(rows)

    title = f"#### {system_name} results for: '{query}'"
    display(Markdown(title))

    display(
        df.style
        .hide(axis="index")
        .format({"Score": "{:.3f}"})
        .set_properties(subset=["Title", "Info", "Snippet"], **{"text-align"
```

```
        .set_properties(subset=["Snippet"], **{"white-space": "pre-wrap"})
    )

    return df
```

1. Data exploration and preprocessing

In [4]:
```
# Load small subsets (200 rows)
meta_path = "../data/raw/meta_Appliances.jsonl.gz"
review_path = "../data/raw/Appliances.jsonl.gz"

meta_records = load_jsonl_gz(meta_path, n=200)
review_records = load_jsonl_gz(review_path, n=200)

print(f"Loaded {len(meta_records)} metadata records")
print(f"Loaded {len(review_records)} review records")

# Convert to DataFrames for easier EDA
meta_df = pd.DataFrame(meta_records)
review_df = pd.DataFrame(review_records)

meta_df.head()
```

```
Loaded 200 metadata records
Loaded 200 review records
```

Out[4]:

	main_category	title	average_rating	rating_number	features	
0	Industrial & Scientific	ROVSUN Ice Maker Machine Countertop, Make 44lb...	3.7	61	[【Quick Ice Making】This countertop ice machine...	
1	Tools & Home Improvement	HANSGO Egg Holder for Refrigerator, Deviled Eg...	4.2	75	[Plastic, Practical Kitchen Storage - Our egg ...	
2	Tools & Home Improvement	Clothes Dryer Drum Slide, General Electric, Ho...	3.5	18	[]	[Brar drum sl
3	Tools & Home Improvement	154567702 Dishwasher Lower Wash Arm Assembly f...	4.5	26	[MODEL NUMBER:154567702 Dishwasher Lower Wash ...	NUMBEI Dishw
4	Tools & Home Improvement	Whirlpool W10918546 Igniter	3.8	12	[This is a Genuine OEM Replacement Part.]	[Whirl

1.1. Dataset overview

In [5]:

```
print("Metadata fields:", meta_df.columns.tolist())
print("Review fields:", review_df.columns.tolist())
print()
print("Metadata shape:", meta_df.shape)
print("Review shape:", review_df.shape)
```

```
Metadata fields: ['main_category', 'title', 'average_rating', 'rating_numbe
r', 'features', 'description', 'price', 'images', 'videos', 'store', 'catego
ries', 'details', 'parent_asin', 'bought_together']
Review fields: ['rating', 'title', 'text', 'images', 'asin', 'parent_asin',
'user_id', 'timestamp', 'helpful_vote', 'verified_purchase']

Metadata shape: (200, 14)
Review shape: (200, 10)
```

1.2. Sample records

In [6]:

```
meta_df.sample(3)
review_df.sample(3)
```

Out[6]:

	rating	title	text	images	asin	parent_asin	
92	1.0	One Star	It does not fit the Haier WD-2800-25	[]	B001RCM6KW	B001RCM6KW	AG7WKTZINOFI
169	5.0	Highly recommend	We have a whirlpool and this worked perfectly....	[]	B06XFWNT3M	B0B96NWZ9J	AFGIT2Y7X5GIN
121	5.0	A+	Worked great	[]	B00AYBKUAA	B00AYBKUAA	AFUMP7GO4PZ3S

1.3. Data preprocessing

In [7]:

```
# Fields we decided to use for retrieval
META_TEXT_FIELDS = ["title", "description", "features"]
REVIEW_TEXT_FIELDS = ["title", "text"]

documents = []

# Process metadata entries
for _, row in meta_df.iterrows():
    text_parts = []

    for field in META_TEXT_FIELDS:
        value = row.get(field)
        if isinstance(value, str) and len(value.strip()) > 0:
            text_parts.append(value.strip())

    # Skip if no meaningful text
    if len(text_parts) == 0:
        continue

    full_text = "\n\n".join(text_parts)

    documents.append(
        Document(
            page_content=full_text,
            metadata={
                "source": "metadata",
                "asin": row.get("parent_asin"),
                "main_category": row.get("main_category"),
                "categories": row.get("categories"),
            }
        )
    )

# Process review entries
for _, row in review_df.iterrows():
    text_parts = []
```

```
    for field in REVIEW_TEXT_FIELDS:
        value = row.get(field)
        if isinstance(value, str) and len(value.strip()) > 0:
            text_parts.append(value.strip())

    if len(text_parts) == 0:
        continue

    full_text = "\n\n".join(text_parts)

    documents.append(
        Document(
            page_content=full_text,
            metadata={
                "source": "review",
                "asin": row.get("asin"),
                "rating": row.get("rating"),
                "verified_purchase": row.get("verified_purchase"),
            }
        )
    )

len(documents)
```

Out[7]: 399

1.3.1 Selection and justification of fields for retrieval

For retrieval, we focused on fields that contain meaningful natural-language text.

- From the metadata, `title`, `description`, and sometimes `features` provide the most useful product information.
- From the reviews, the key fields are `text` (full review body) and `title` (short summary).
  Other fields like price, images, or ratings don't add much value for text-based retrieval.

1.3.2. Description of text preprocessing

Light preprocessing was applied to keep the text clean but still semantically rich:

- all text lowercased
- extra whitespace and URLs removed
- punctuation kept
- empty or extremely short entries removed

There is no stemming or lemmatization involved, as modern embedding models handle them well on their own.

2. Keyword-based retrieval system (BM25)

In [8]:
```
bm25, tokenized = build_bm25(documents)
save_bm25(bm25, tokenized, save_dir="../bm25_index/")

bm25, tokenized = load_bm25("../bm25_index/")
```

2.1. Example query results

In [9]:
```
query = "quiet dishwasher stainless steel"
results = bm25_search(query, bm25, documents, k=5)

display_search_results(results, query=query, system_name="BM25 search");
```

BM25 search results for: 'quiet dishwasher stainless steel'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	11.088	metadata	B0052EK0MW	KitchenAid Superba Series KUDS30SXSS Fully Integrated Dishwasher, Stainless Steel	Appliances | Appliances > Dishwashers > Built-In Dishwashers	KitchenAid Superba Series KUDS30SXSS Fully Integrated Dishwasher, Stainless Steel
2	9.758	metadata	B09XXFX5SK	MLGB Stainless Steel Brushed Pattern Dishwasher Magnet Cover Panel Decal Home Appliance Art, Stainless Steel Fridge Door Cover Decals Magnetic, Black, Mobile Magnetic 23" x 26"	Amazon Home | Appliances > Parts & Accessories > Dishwasher Parts & Accessories > Hoses	MLGB Stainless Steel Brushed Pattern Dishwasher Magnet Cover Panel Decal Home Appliance Art, Stainless Steel Fridge Door Cover Decals Magnetic, Black, Mobile Magnetic 23" x 26"
3	7.350	metadata	B00PLD0YUW	midea HS-209BESS Beer/Beverage Refrigerator and Dispenser, 5.7 Cubic Feet, Stainless Steel	Appliances | Appliances > Refrigerators, Freezers & Ice Makers > Kegerators	midea HS-209BESS Beer/Beverage Refrigerator and Dispenser, 5.7 Cubic Feet, Stainless Steel
4	7.242	review	B076VPHFH6	stainless steel pitcher	rating=5.0, verified=True	I actually purchased this to pour one cup of water into my individual cup coffee maker. So I don't froth milk.. but this pitcher is good for my average mug, I purchased a larger one for my larger...
5	7.143	metadata	B005H47EJ4	GE CGS985SETSS Cafe 30" Stainless Steel Gas Sealed Burner Range - Convection	Appliances | Appliances > Ranges, Ovens & Cooktops > Ranges > Freestanding Ranges	GE CGS985SETSS Cafe 30" Stainless Steel Gas Sealed Burner Range - Convection

3. Semantic-based retrival system

3.1. Embeddings with sentence transformers

In [10]:
```
embeddings, model = build_embeddings(documents)
index = build_faiss_index(embeddings)
save_faiss(index, documents, save_dir="../semantic_index/")
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set
a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights:   0%|          | 0/103 [00:00<?, ?it/s]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key                     | Status     |  |
------------------------+------------+--+-
embeddings.position_ids | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architectur
e; not ok if you expect identical arch.
Batches:   0%|          | 0/13 [00:00<?, ?it/s]
```

3.2. FAISS index and documents

In [11]:
```
index, documents = load_faiss("../semantic_index/")
```

3.3. Example query results

In [12]:
```
query = "quiet dishwasher stainless steel"
results = semantic_search(query, index, model, documents, k=5)

display_search_results(results, query=query, system_name="Semantic search");
```

Semantic search results for: 'quiet dishwasher stainless steel'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	0.773	metadata	B0052EK0MW	KitchenAid Superba Series KUDS30SXSS Fully Integrated Dishwasher, Stainless Steel	Appliances | Appliances > Dishwashers > Built-In Dishwashers	KitchenAid Superba Series KUDS30SXSS Fully Integrated Dishwasher, Stainless Steel
2	0.954	review	B09LM3ZW5N	Easy to clean, look good, love these	rating=5.0, verified=False	Before silicon counter gap covers came out I had thinner Medal ones I had to paint black, and they didn't clean well! And Those made noise, and the paint would rub off, and I had to paint them...
3	0.957	metadata	B00T5K1924	Frigidaire Professional FPID2495QF Fully Integrated Dishwasher	Appliances > Dishwashers > Built-In Dishwashers	Frigidaire Professional FPID2495QF Fully Integrated Dishwasher
4	0.968	review	B0047PF5MM	Finally, the branded product to clean my Miele!	rating=5.0, verified=True	I have been trying to find something that cleans my 6 year old dish washer and in the past it was run the machine empty. Although we have never noticed any deterioration in the wonderful cleaning...
5	0.989	metadata	B01HP14VVU	ERP 809006501 Dishwasher Lower Seal	Tools & Home Improvement | Appliances > Parts & Accessories	ERP 809006501 Dishwasher Lower Seal

4. Qualitative evaluation

In [13]:
```
queries = [
    "stainless steel dishwasher",
    "refrigerator water filter",
    "gas range 30 inch",
    "something to keep drinks cold in a dorm room",
    "appliance for washing dishes quietly at night",
    "replacement part to stop a dishwasher from leaking at the bottom",
```

```
    "small machine for drying clothes in an apartment",
    "best dishwasher for a small apartment under $800",
    "energy-efficient refrigerator for a family of four",
    "reliable stove for frequent home cooking that is easy to clean",
]
```

In [14]:
```
k = 5

for query in queries:
    bm25_results = bm25_search(query, bm25, documents, k=k)
    semantic_results = semantic_search(query, index, model, documents, k=k)

    bm25_df = display_search_results(
        bm25_results, query=query, system_name="BM25")

    semantic_df = display_search_results(
        semantic_results, query=query, system_name="Semantic")
```

BM25 results for: 'stainless steel dishwasher'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	11.088	metadata	B0052EK0MW	KitchenAid Superba Series KUDS30SXSS Fully Integrated Dishwasher, Stainless Steel	Appliances | Appliances > Dishwashers > Built-In Dishwashers	KitchenAid Superba Series KUDS30SXSS Fully Integrated Dishwasher, Stainless Steel
2	9.758	metadata	B09XXFX5SK	MLGB Stainless Steel Brushed Pattern Dishwasher Magnet Cover Panel Decal Home Appliance Art, Stainless Steel Fridge Door Cover Decals Magnetic, Black, Mobile Magnetic 23" x 26"	Amazon Home | Appliances > Parts & Accessories > Dishwasher Parts & Accessories > Hoses	MLGB Stainless Steel Brushed Pattern Dishwasher Magnet Cover Panel Decal Home Appliance Art, Stainless Steel Fridge Door Cover Decals Magnetic, Black, Mobile Magnetic 23" x 26"
3	7.350	metadata	B00PLD0YUW	midea HS-209BESS Beer/Beverage Refrigerator and Dispenser, 5.7 Cubic Feet, Stainless Steel	Appliances | Appliances > Refrigerators, Freezers & Ice Makers > Kegerators	midea HS-209BESS Beer/Beverage Refrigerator and Dispenser, 5.7 Cubic Feet, Stainless Steel
4	7.242	review	B076VPHFH6	stainless steel pitcher	rating=5.0, verified=True	I actually purchased this to pour one cup of water into my individual cup coffee maker. So I don't froth milk.. but this pitcher is good for my average mug, I purchased a larger one for my larger...
5	7.143	metadata	B005H47EJ4	GE CGS985SETSS Cafe 30" Stainless Steel Gas Sealed Burner Range - Convection	Appliances | Appliances > Ranges, Ovens & Cooktops > Ranges > Freestanding Ranges	GE CGS985SETSS Cafe 30" Stainless Steel Gas Sealed Burner Range - Convection

Semantic results for: 'stainless steel dishwasher'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	0.623	metadata	B0052EK0MW	KitchenAid Superba Series KUDS30SXSS Fully Integrated Dishwasher, Stainless Steel	Appliances | Appliances > Dishwashers > Built-In Dishwashers	KitchenAid Superba Series KUDS30SXSS Fully Integrated Dishwasher, Stainless Steel
2	0.801	metadata	B00T5K1924	Frigidaire Professional FPID2495QF Fully Integrated Dishwasher	Appliances > Dishwashers > Built-In Dishwashers	Frigidaire Professional FPID2495QF Fully Integrated Dishwasher
3	0.853	metadata	B01HP14VVU	ERP 809006501 Dishwasher Lower Seal	Tools & Home Improvement | Appliances > Parts & Accessories	ERP 809006501 Dishwasher Lower Seal
4	0.892	review	B0047PF5MM	Finally, the branded product to clean my Miele!	rating=5.0, verified=True	I have been trying to find something that cleans my 6 year old dish washer and in the past it was run the machine empty. Although we have never noticed any deterioration in the wonderful cleaning...
5	0.900	metadata	B089YTWKC3	Dishwasher Silverware Basket, Genuine Original Equipment Manufacturer Dishwasher Cutlery Basket Compatible with Kenmore, Whirlpool, Bosch, Maytag, KitchenAid, Samsung, GE, and more	Appliances | Appliances > Parts & Accessories > Dishwasher Parts & Accessories > Baskets	Dishwasher Silverware Basket, Genuine Original Equipment Manufacturer Dishwasher Cutlery Basket Compatible with Kenmore, Whirlpool, Bosch, Maytag, KitchenAid, Samsung, GE, and more

BM25 results for: 'refrigerator water filter'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	8.264	metadata	B00YD32MBK	Replacement for LG LFX28978ST/00 Refrigerator Water Filter - Compatible with LG LFX28978ST/00 Fridge Water Filter Cartridge	Tools & Home Improvement | Appliances > Parts & Accessories > Refrigerator Parts & Accessories > Water Filters	Replacement for LG LFX28978ST/00 Refrigerator Water Filter - Compatible with LG LFX28978ST/00 Fridge Water Filter Cartridge
2	8.124	metadata	B0BG2S3W93	Crystala Filters LT1000PC Refrigerator Water Filter, Water Filter ADQ747935 Compatible with LT1000PC, LT1000PC/PCS, LT-1000PC, MDJ64844601, ADQ747935, ADQ74793504 Water Filter (4 Pack)	Amazon Home | Appliances > Parts & Accessories > Refrigerator Parts & Accessories > Water Filters	Crystala Filters LT1000PC Refrigerator Water Filter, Water Filter ADQ747935 Compatible with LT1000PC, LT1000PC/PCS, LT-1000PC, MDJ64844601, ADQ747935, ADQ74793504 Water Filter (4 Pack)
3	7.940	metadata	B00YD33QME	4-Pack Replacement for LG LFX25974SW Refrigerator Water Filter - Compatible with LG 5231JA2002A, LT500P Fridge Water Filter Cartridge	Tools & Home Improvement | Appliances > Parts & Accessories > Refrigerator Parts & Accessories > Water Filters	4-Pack Replacement for LG LFX25974SW Refrigerator Water Filter - Compatible with LG 5231JA2002A, LT500P Fridge Water Filter Cartridge
4	7.788	metadata	B07CPY3X2X	Compatible with LT700P Refrigerator Water Filter, LG Water Filter Compatible with LT700P, ADQ36006101, ADQ36006102 and KENMORE 9690, 469690-3 Pack	Tools & Home Improvement | Appliances > Parts & Accessories > Refrigerator Parts & Accessories > Water Filters	Compatible with LT700P Refrigerator Water Filter, LG Water Filter Compatible with LT700P, ADQ36006101, ADQ36006102 and KENMORE 9690, 469690-3 Pack
5	7.788	metadata	B00YD3JBAK	3-Pack Replacement for KitchenAid KSSC36FMS Refrigerator Water Filter - Compatible with	Amazon Home | Appliances > Parts & Accessories > Refrigerator Parts &	3-Pack Replacement for KitchenAid KSSC36FMS Refrigerator Water Filter - Compatible with

Rank	Score	Source	ASIN	Title	Info	Snippet
				KitchenAid 4396508, 4396509, 4396510 Fridge Water Filter Cartridge	Accessories > Water Filters	KitchenAid 4396508, 4396509, 4396510 Fridge Water Filter Cartridge

Semantic results for: 'refrigerator water filter'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	0.430	review	B000UW2DTE	Wirlpool Water filter for Refrigerater	rating=5.0, verified=True	This is my second refrigerater with a waterfilter on the bottom and I love it. The water tastes teriffic and you do not need to buy water in the store. I recomment Wirlpool refrigerator hightly...
2	0.663	metadata	B00UTC2YOY	EveryDrop by Whirlpool Refrigerator Water Filter 3 (Pack of 3)	Tools & Home Improvement | Appliances > Parts & Accessories > Refrigerator Parts & Accessories > Water Filters	EveryDrop by Whirlpool Refrigerator Water Filter 3 (Pack of 3)
3	0.706	review	B0045LLC7K	Original Frigidaire water filter that is easy to install	rating=5.0, verified=True	This works in my Frigidaire Model J51-23, so what's not to love. It is exactly the same filter as the one that came with the fridge. Also, it was easy to install. I removed my old one by pressing...
4	0.714	metadata	B018K8UZL2	NEW GE Smart Water Filter BY-PASS Fitting Plug WR02X11705 Refrigerator MWF GFW	Amazon Home | Appliances > Parts & Accessories > Refrigerator Parts & Accessories	NEW GE Smart Water Filter BY-PASS Fitting Plug WR02X11705 Refrigerator MWF GFW
5	0.735	metadata	B01CULFN4Y	Aqua Fresh RPWF Refrigerator Water Filter Compatible with GE RPWF, RWF1063, RWF3600A, WSG-4, DWF-36, R-3600, MPF15350,	Amazon Home | Appliances > Parts & Accessories > Refrigerator Parts & Accessories > Water Filters	Aqua Fresh RPWF Refrigerator Water Filter Compatible with GE RPWF, RWF1063, RWF3600A, WSG-4, DWF-36, R-3600, MPF15350, OPFG3-RF300 (3 Pack)

Rank	Score	Source	ASIN	Title	Info	Snippet
				OPFG3-RF300 (3 Pack)		

BM25 results for: 'gas range 30 inch'

Rank	Score	Source	ASIN	Title	Info	
1	13.003	metadata	B005H47EJ4	GE CGS985SETSS Cafe 30" Stainless Steel Gas Sealed Burner Range - Convection	Appliances | Appliances > Ranges, Ovens & Cooktops > Ranges > Freestanding Ranges	GE CGS985SETSS 30" Stainless Stee Sealed Burner Ra Convection
2	8.879	metadata	B00SSZO6P4	VINTAGE Frigidaire Range 5306590805 LARGE 8 INCH BURNER COMPLIANT BLACK	Appliances | Appliances > Parts & Accessories > Range Parts & Accessories > Accessories	VINTAGE Frigidai Range 53065908 LARGE 8 INCH BU COMPLIANT BLA
3	8.809	metadata	B09P9XS95N	Stove Burner Covers for Samsung Gas Range | Gas Stove Protector Set with 2 Pack Stove Gap Covers, 1Pack Rag | Gas Stove Liners, Reusable	Tools & Home Improvement | Appliances > Parts & Accessories > Range Parts & Accessories > Accessories	Stove Burner Cov Samsung Gas Ra Gas Stove Protect with 2 Pack Stove Covers, 1Pack Ra Stove Liners, Reu
4	8.776	review	B09G76121T	Samsung perfect fit.	rating=5.0, verified=True	We just had a Sa gas range installe kept spilling thin ordered this liner. arrived today and perfect. I am ecst more worrying ab overspills on the Samsung Gas...
5	8.731	metadata	B088WFND8V	8 PCS Gas Stove Burner Covers,Black and Silver Gas Range Protectors,Reusable,Non-Stick Heat-Resistant Covers,Size 10.6 X 10.6 Inches	Tools & Home Improvement | Appliances > Parts & Accessories > Range Parts & Accessories > Accessories	8 PCS Gas Stove E Covers,Black and Gas Range Protectors,Reusab Stick Heat-Resista Covers,Size 10.6 X Inches

Semantic results for: 'gas range 30 inch'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	1.001	metadata	B005H47EJ4	GE CGS985SETSS Cafe 30" Stainless Steel Gas Sealed Burner Range - Convection	Appliances | Appliances > Ranges, Ovens & Cooktops > Ranges > Freestanding Ranges	GE CGS985SETSS Cafe 30" Stainless Steel Gas Sealed Burner Range - Convection
2	1.213	metadata	B078QB5M4B	Compatible Oven or Broiler Igniter for General Electric JGBP32WEJ3WW, General Electric RGB745WEH7WW, General Electric JGBP28BEA6WH, Roper F8958W0 Gas Range	Tools & Home Improvement | Appliances > Parts & Accessories	Compatible Oven or Broiler Igniter for General Electric JGBP32WEJ3WW, General Electric RGB745WEH7WW, General Electric JGBP28BEA6WH, Roper F8958W0 Gas Range
3	1.238	metadata	B07FNSTXCM	Kenmore 30" Freestanding Electric Range with 4.8 Cubic Ft. Total Capacity, Stainless Steel	Appliances | Appliances > Ranges, Ovens & Cooktops > Ranges > Freestanding Ranges	Kenmore 30" Freestanding Electric Range with 4.8 Cubic Ft. Total Capacity, Stainless Steel
4	1.243	metadata	B00SSZO6P4	VINTAGE Frigidaire Range 5306590805 LARGE 8 INCH BURNER COMPLIANT BLACK	Appliances | Appliances > Parts & Accessories > Range Parts & Accessories > Accessories	VINTAGE Frigidaire Range 5306590805 LARGE 8 INCH BURNER COMPLIANT BLACK
5	1.267	metadata	B09P9XS95N	Stove Burner Covers for Samsung Gas Range | Gas Stove Protector Set with 2 Pack Stove Gap Covers, 1Pack Rag | Gas Stove Liners, Reusable	Tools & Home Improvement | Appliances > Parts & Accessories > Range Parts & Accessories > Accessories	Stove Burner Covers for Samsung Gas Range | Gas Stove Protector Set with 2 Pack Stove Gap Covers, 1Pack Rag | Gas Stove Liners, Reusable

BM25 results for: 'something to keep drinks cold in a dorm room'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	12.242	review	B08ZYJ8CRX	Love to use a lot everyday!!	rating=5.0, verified=True	Family love it and use for cold drinks !!
2	7.173	review	B087WK8SJH	Works Well	rating=5.0, verified=False	I love having a refrigerator just for beer! We have half size refrigerator for drinks, but that gets full pretty quickly with water bottles, sodas, and juice. Now we can put the beer and malt...
3	5.141	review	B082FBV7XY	Great product	rating=5.0, verified=True	Helps keep my kitchen clean
4	4.829	metadata	B0B1TXLLY7	Compact Laundry Dryer, 1400W Electric Portable Clothes Dryer, 9LBS Laundry Dryer, Premium Compact Tumbler Stainless Steel Laundry Dryer with Exhaust Pipe, LCD Panel Dryer for Apartments, Home, Dorm (Whit-9LBs)	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Dryers	Compact Laundry Dryer, 1400W Electric Portable Clothes Dryer, 9LBS Laundry Dryer, Premium Compact Tumbler Stainless Steel Laundry Dryer with Exhaust Pipe, LCD Panel Dryer for Apartments, Home, Dorm...
5	4.692	metadata	B07VVR2TNH	Repairwares Washing Machine Cold Water Inlet Valve Assembly 422244 00422244 1105556 WV2244 PS3462925 PS8713229 for Select Bosch Washer Models	Tools & Home Improvement | Appliances > Parts & Accessories > Washer Parts & Accessories > Doors	Repairwares Washing Machine Cold Water Inlet Valve Assembly 422244 00422244 1105556 WV2244 PS3462925 PS8713229 for Select Bosch Washer Models

Semantic results for: 'something to keep drinks cold in a dorm room'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	1.140	review	B08ZYJ8CRX	Love to use a lot everyday!!	rating=5.0, verified=True	Family love it and use for cold drinks !!
2	1.170	review	B07YF9SGBW	Pellet Ice!!	rating=5.0, verified=True	If you love pellet ice then this is the unit for you. I put this in my bar and use it almost every day. It does put out some heat and is a little loud when it is making ice but I absolutely love it!
3	1.199	metadata	B094MYFT1N	NewAir NBC126SS02 Beverage Refrigerator and Cooler, Holds up to 126 Cans, Cools Down to 37 Degrees Perfect for Beer Wine or Soda, 126 Can, Silver, 126 Can (Renewed)	Appliances | Appliances > Refrigerators, Freezers & Ice Makers > Beverage Refrigerators	NewAir NBC126SS02 Beverage Refrigerator and Cooler, Holds up to 126 Cans, Cools Down to 37 Degrees Perfect for Beer Wine or Soda, 126 Can, Silver, 126 Can (Renewed)
4	1.207	metadata	B083Q6Y54F	G.a HOMEFAVOR Cold Brew Coffee Infuser 64oz (2 Quart), Stainless Steel Filter Kit for Wide Mason Jar and Iced Tea Maker at Home	Amazon Home | Small Appliance Parts & Accessories > Coffee & Espresso Machine Parts & Accessories > Coffee Machine Accessories > Coffee Filters	G.a HOMEFAVOR Cold Brew Coffee Infuser 64oz (2 Quart), Stainless Steel Filter Kit for Wide Mason Jar and Iced Tea Maker at Home

Rank	Score	Source	ASIN	Title	Info	Snippet
5	1.249	review	B087WK8SJH	Works Well	rating=5.0, verified=False	I love having a refrigerator just for beer! We have half size refrigerator for drinks, but that gets full pretty quickly with water bottles, sodas, and juice. Now we can put the beer and malt...

BM25 results for: 'appliance for washing dishes quietly at night'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	5.179	review	B001PIAPP6	Great deal	rating=5.0, verified=True	My local appliance shop wanted $45.20 for this part. This works just as well as his part would have and both parts were Not OEM
2	4.595	metadata	B0053F9LCK	GENUINE Whirlpool 8183202 Hinge for Washing Machine	Tools & Home Improvement | Appliances > Parts & Accessories > Washer Parts & Accessories	GENUINE Whirlpool 8183202 Hinge for Washing Machine
3	4.571	metadata	B09B3TZ6KK	ASSINAI Portable Mini Washing Machine, Ultrasonic Washing Machine 3 In 1 Dishwasher Mini Light, Convenient for Travel, Family Business Travel, USB	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Washers	ASSINAI Portable Mini Washing Machine, Ultrasonic Washing Machine 3 In 1 Dishwasher Mini Light, Convenient for Travel, Family Business Travel, USB
4	4.449	metadata	B00K321G0S	Samsung WF45H6300AW + DV45H6300EW with SuperSpeed and PowerFoam washing technologies!	Appliances	Samsung WF45H6300AW + DV45H6300EW with SuperSpeed and PowerFoam washing technologies!
5	4.312	metadata	B08CNMM3L5	ClimaTek Washing Machine Drain Pump Fits Maytag W10727777	Tools & Home Improvement | Appliances > Parts & Accessories > Washer Parts & Accessories > Drain Pumps	ClimaTek Washing Machine Drain Pump Fits Maytag W10727777

Semantic results for: 'appliance for washing dishes quietly at night'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	1.049	metadata	B09B3TZ6KK	ASSINAI Portable Mini Washing Machine, Ultrasonic Washing Machine 3 In 1 Dishwasher Mini Light, Convenient for Travel, Family Business Travel, USB	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Washers	ASSINAI Portable Mini Washing Machine, Ultrasonic Washing Machine 3 In 1 Dishwasher Mini Light, Convenient for Travel, Family Business Travel, USB
2	1.129	review	B089YD3Z91	Rinses dishes....lovely,. could have saved more than the exorbitant price PLUS the pai	rating=2.0, verified=True	Not worth the price. Takes up too much space, putting water in for the wash cycle is difficult AND requires a space to drain the "wash water" into unless willing to hook it up to your kitchen sink,...
3	1.182	metadata	B09N8NXR35	The Laundry Alternative PuriFI - Portable Small Washer for Apartment & RV - Combo Diaper Washer & Automatic Clothes Washing Machine - Best Rated Quiet, Compact, Best Rated Small Washing Machine	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Washers	The Laundry Alternative PuriFI - Portable Small Washer for Apartment & RV - Combo Diaper Washer & Automatic Clothes Washing Machine - Best Rated Quiet, Compact, Best Rated Small Washing Machine
4	1.194	metadata	B08M5XJQYV	Xiaqing Clothes Dryer Portable Fast 1000W Dryer Machine,Portable Dryer for Apartments,New Generation Electric Clothes Drying	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Dryers	Xiaqing Clothes Dryer Portable Fast 1000W Dryer Machine,Portable Dryer for Apartments,New Generation Electric Clothes Drying
5	1.222	review	B0047PF5MM	Finally, the branded product to clean my Miele!	rating=5.0, verified=True	I have been trying to find something that cleans my 6 year old dish washer and in the past it was run the machine empty. Although we have

Rank	Score	Source	ASIN	Title	Info	Snippet
						never noticed any deterioration in the wonderful cleaning...

BM25 results for: 'replacement part to stop a dishwasher from leaking at the bottom'

Rank	Score	Source	ASIN	Title	Info	Sn
1	9.353	review	B08QTQW9VQ	It leaking on bottom but those company is out of business so can't replace!!	rating=3.0, verified=True	Love it but can't re because of compa out of business so brought different and good so far
2	8.085	metadata	B08ZSN1RRB	HASMX W11157084 WP8561996 Dishwasher Upper Rack Wheel Mount Replacement Part for Whirlpool Maytag - Replaces Part Numbers W11157084, AP6285708, WP8561996, 8561996, PS973972, B0050O2HIO, B00JLMM1V4 (2)	Tools & Home Improvement | Appliances > Parts & Accessories	HASMX W111570 WP8561996 Dishw Upper Rack Whee Mount Replaceme Part for Whirlpool Maytag - Replaces Numbers W11157 AP6285708, WP8561996, 8561 PS973972, B0050O2HIO, B00JLMM1V4 (2)
3	7.986	metadata	B0C7K98JZS	154567702 Dishwasher Lower Wash Arm Assembly for Kenmore Electrolux Dishwasher Bottom Lower Spray Arm 5304518927, AP6810011,154567701...	Tools & Home Improvement | Appliances > Parts & Accessories > Dryer Parts & Accessories > Replacement Parts	154567702 Dishwa Lower Wash Arm Assembly for Kenr Electrolux Dishwas Bottom Lower Spr Arm 5304518927, AP6810011,15456
4	6.914	review	B01M28A97G	Works well	rating=5.0, verified=True	Perfect replaceme part for my dryer.
5	6.527	review	B08LVFQYLT	This holds 14 eggs nicely in the fridge.	rating=4.0, verified=False	With 2 containers can store 28 eggs. These wash easily, not in the dishwas The bottom is ok, the top warped.

Semantic results for: 'replacement part to stop a dishwasher from leaking at the bottom'

Rank	Score	Source	ASIN	Title	Info	Sn
1	0.743	review	B08QTQW9VQ	It leaking on bottom but those company is out of business so can’t replace!!	rating=3.0, verified=True	Love it but can’t re because of compa out of business so brought different c and good so far
2	0.859	metadata	B01HP14VVU	ERP 809006501 Dishwasher Lower Seal	Tools & Home Improvement | Appliances > Parts & Accessories	ERP 809006501 Dishwasher Lower
3	0.914	review	B09DVH6MN7	Works fits our dishwasher	rating=5.0, verified=True	Fits
4	0.987	metadata	B00T5K1924	Frigidaire Professional FPID2495QF Fully Integrated Dishwasher	Appliances > Dishwashers > Built-In Dishwashers	Frigidaire Professic FPID2495QF Fully Integrated Dishwas
5	1.012	metadata	B0C7K98JZS	154567702 Dishwasher Lower Wash Arm Assembly for Kenmore Electrolux Dishwasher Bottom Lower Spray Arm 5304518927, AP6810011,154567701…	Tools & Home Improvement | Appliances > Parts & Accessories > Dryer Parts & Accessories > Replacement Parts	154567702 Dishwa Lower Wash Arm Assembly for Kenn Electrolux Dishwas Bottom Lower Spra Arm 5304518927, AP6810011,154567

BM25 results for: 'small machine for drying clothes in an apartment'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	15.838	metadata	B09N8NXR35	The Laundry Alternative PuriFI - Portable Small Washer for Apartment & RV - Combo Diaper Washer & Automatic Clothes Washing Machine - Best Rated Quiet, Compact, Best Rated Small Washing Machine	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Washers	The Laundry Alternative PuriFI - Portable Small Washer for Apartment & RV - Combo Diaper Washer & Automatic Clothes Washing Machine - Best Rated Quiet, Compact, Best Rated Small Washing Machine
2	11.734	metadata	B08M5XJQYV	Xiaqing Clothes Dryer Portable Fast 1000W Dryer Machine,Portable Dryer for Apartments,New Generation Electric Clothes Drying	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Dryers	Xiaqing Clothes Dryer Portable Fast 1000W Dryer Machine,Portable Dryer for Apartments,New Generation Electric Clothes Drying
3	10.678	review	B095M5PJHS	Washes Good	rating=5.0, verified=True	This washes clothes just as good as any washing machine. Only issues I have is it has the smallest drain hose and takes forever to drain. The spin cycle is a joke. Don’t buy it for the spin cycle...
4	6.636	review	B07FTXG5SG	Quick and easy	rating=4.0, verified=True	Love the convenience of doing laundry inside my apartment, but do have a problem attaching the drain pipe when ready to drain the tub. Trial and error, I'm getting there. I have done several loads...
5	5.619	review	B07F618LKC	Perfect addition to my apartment !	rating=5.0, verified=True	It arrived today, well packaged, and I've used it once. Very pleased

Rank	Score	Source	ASIN	Title	Info	Snippet
						with this compact, quiet dryer !

Semantic results for: 'small machine for drying clothes in an apartment'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	0.491	metadata	B08M5XJQYV	Xiaqing Clothes Dryer Portable Fast 1000W Dryer Machine,Portable Dryer for Apartments,New Generation Electric Clothes Drying	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Dryers	Xiaqing Clothes Dryer Portable Fast 1000W Dryer Machine,Portable Dryer for Apartments,New Generation Electric Clothes Drying
2	0.829	metadata	B09N8NXR35	The Laundry Alternative PuriFI - Portable Small Washer for Apartment & RV - Combo Diaper Washer & Automatic Clothes Washing Machine - Best Rated Quiet, Compact, Best Rated Small Washing Machine	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Washers	The Laundry Alternative PuriFI - Portable Small Washer for Apartment & RV - Combo Diaper Washer & Automatic Clothes Washing Machine - Best Rated Quiet, Compact, Best Rated Small Washing Machine
3	0.899	metadata	B09B3TZ6KK	ASSINAI Portable Mini Washing Machine, Ultrasonic Washing Machine 3 In 1 Dishwasher Mini Light, Convenient for Travel, Family Business Travel, USB	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Washers	ASSINAI Portable Mini Washing Machine, Ultrasonic Washing Machine 3 In 1 Dishwasher Mini Light, Convenient for Travel, Family Business Travel, USB
4	0.952	metadata	B01IDSLR7K	XtremepowerUS Portable Ventless Clothes Dryer, Dryer Rack w/Heater	Appliances	XtremepowerUS Portable Ventless Clothes Dryer, Dryer Rack w/Heater
5	0.975	review	B095M5PJHS	Washes Good	rating=5.0, verified=True	This washes clothes just as good as any washing machine. Only issues I have is it has the smallest drain hose and takes forever to drain. The spin cycle is a joke. Don't buy it for the spin cycle...

BM25 results for: 'best dishwasher for a small apartment under $800'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	13.997	metadata	B09N8NXR35	The Laundry Alternative PuriFI - Portable Small Washer for Apartment & RV - Combo Diaper Washer & Automatic Clothes Washing Machine - Best Rated Quiet, Compact, Best Rated Small Washing Machine	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Washers	The Laundry Alternative PuriFI - Portable Small Washer for Apartment & RV - Combo Diaper Washer & Automatic Clothes Washing Machine - Best Rated Quiet, Compact, Best Rated Small Washing Machine
2	5.783	review	B001TH7GZA	They don't fit properly	rating=2.0, verified=True	Not the best quality
3	5.619	review	B07F618LKC	Perfect addition to my apartment !	rating=5.0, verified=True	It arrived today, well packaged, and I've used it once. Very pleased with this compact, quiet dryer !
4	5.307	review	B000LC84PK	For small eggs	rating=2.0, verified=True	I'm afraid to put eggs in this thing it's made of hard plastic and the holes it has are to small for my large eggs .
5	4.806	review	B0052KUJKY	Best filter for good coffee.	rating=5.0, verified=True	Unbleached filters always make the coffee taste better.

Semantic results for: 'best dishwasher for a small apartment under $800'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	0.959	metadata	B00T5K1924	Frigidaire Professional FPID2495QF Fully Integrated Dishwasher	Appliances > Dishwashers > Built-In Dishwashers	Frigidaire Professional FPID2495QF Fully Integrated Dishwasher
2	0.962	review	B089YD3Z91	Rinses dishes....lovely,. could have saved more than the exorbitant price PLUS the pai	rating=2.0, verified=True	Not worth the price. Takes up too much space, putting water in for the wash cycle is difficult AND requires a space to drain the "wash water" into unless willing to hook it up to your kitchen sink,...
3	0.967	review	B0047PF5MM	Finally, the branded product to clean my Miele!	rating=5.0, verified=True	I have been trying to find something that cleans my 6 year old dish washer and in the past it was run the machine empty. Although we have never noticed any deterioration in the wonderful cleaning...
4	0.974	review	B09LM3ZW5N	Easy to clean, look good, love these	rating=5.0, verified=False	Before silicon counter gap covers came out I had thinner Medal ones I had to paint black, and they didn't clean well! And Those made noise, and the paint would rub off, and I had to paint them...
5	1.003	metadata	B09B3TZ6KK	ASSINAI Portable Mini Washing Machine, Ultrasonic Washing Machine 3 In 1 Dishwasher	Appliances | Appliances > Laundry Appliances > Washers & Dryers >	ASSINAI Portable Mini Washing Machine, Ultrasonic Washing Machine 3 In 1 Dishwasher

Rank	Score	Source	ASIN	Title	Info	Snippet
				Mini Light, Convenient for Travel, Family Business Travel, USB	Portable Washers	Mini Light, Convenient for Travel, Family Business Travel, USB

BM25 results for: 'energy-efficient refrigerator for a family of four'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	7.208	review	B001TJ5380	Four Stars	rating=4.0, verified=True	These water filters do the same job as the name brand from Samsung, but they are less expensive.
2	5.682	review	B08ZYJ8CRX	Love to use a lot everyday!!	rating=5.0, verified=True	Family love it and use for cold drinks !!
3	4.717	review	B07FQJ1SZX	I love my little ice maker	rating=5.0, verified=True	Our refrigerator didn't come with an ice maker and trays are a pain and take up too much space. My uncle gifted me his old one to see if I could repair it but it was not meant to be. So I ordered...
4	4.376	metadata	B09B3TZ6KK	ASSINAI Portable Mini Washing Machine, Ultrasonic Washing Machine 3 In 1 Dishwasher Mini Light, Convenient for Travel, Family Business Travel, USB	Appliances | Appliances > Laundry Appliances > Washers & Dryers > Portable Washers	ASSINAI Portable Mini Washing Machine, Ultrasonic Washing Machine 3 In 1 Dishwasher Mini Light, Convenient for Travel, Family Business Travel, USB
5	3.403	metadata	B0C3C3BSJT	WATERMOON Egg Dispenser For Refrigerator - 24 Count Rolling Egg Holder For Refrigerator Egg Storage Container Egg Drawer For Refrigerator Egg Tray For Refrigerator With 4 Spoons,Grey	Tools & Home Improvement | Appliances > Parts & Accessories > Refrigerator Parts & Accessories > Egg Trays	WATERMOON Egg Dispenser For Refrigerator - 24 Count Rolling Egg Holder For Refrigerator Egg Storage Container Egg Drawer For Refrigerator Egg Tray For Refrigerator With 4 Spoons,Grey

Semantic results for: 'energy-efficient refrigerator for a family of four'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	0.928	review	B087WK8SJH	Works Well	rating=5.0, verified=False	I love having a refrigerator just for beer! We have half size refrigerator for drinks, but that gets full pretty quickly with water bottles, sodas, and juice. Now we can put the beer and malt...
2	0.936	metadata	B007ZRHOJQ	Maytag MFX2570AEB Ice2O 25 Cu. Ft. Black French Door Refrigerator - Energy Star	Appliances | Appliances > Refrigerators, Freezers & Ice Makers > Refrigerators	Maytag MFX2570AEB Ice2O 25 Cu. Ft. Black French Door Refrigerator - Energy Star
3	0.937	metadata	B00U7XR54E	GE GSS25GGHWW Side Refrigerator	Appliances | Appliances > Refrigerators, Freezers & Ice Makers > Refrigerators	GE GSS25GGHWW Side Refrigerator
4	0.974	metadata	B00PLD0YUW	midea HS-209BESS Beer/Beverage Refrigerator and Dispenser, 5.7 Cubic Feet, Stainless Steel	Appliances | Appliances > Refrigerators, Freezers & Ice Makers > Kegerators	midea HS-209BESS Beer/Beverage Refrigerator and Dispenser, 5.7 Cubic Feet, Stainless Steel
5	1.077	review	B07TX49L1X	Waste of money!	rating=1.0, verified=False	These are way too small to fit any kind of refrigerator door I've ever seen.

BM25 results for: 'reliable stove for frequent home cooking that is easy to clean'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	10.012	review	B07935N99S	Great item for easy cleanup.	rating=5.0, verified=True	These are great, especially when cooking items that overflow the pot , like soup. When a spill occurs, you simply put the burner in the sink to soak. Minimal mess left on stove. After it soaks and...
2	8.204	review	B00JGM7Z5Q	Great for extra counter space.	rating=4.0, verified=False	This is great for a stove that has raised burners. It gives extra counter space when only using two burners on the stove. Great for small kitchen spaces. Love the bamboo. It's lightweight, durable...
3	7.517	review	B076ZRN68C	Easy, peasy	rating=5.0, verified=True	Everything is great, easy to trim to stove and counter design. Thanks
4	7.137	metadata	B09KH32SSR	Singring Stove Cover Reusable Gas Stove 0.4mm thick Burner Covers Non Stick Stove Protector Easy Cleaning Liners for Gas Ranges	Tools & Home Improvement | Appliances > Parts & Accessories > Range Parts & Accessories > Accessories	Singring Stove Cover Reusable Gas Stove 0.4mm thick Burner Covers Non Stick Stove Protector Easy Cleaning Liners for Gas Ranges
5	7.105	review	B09LM3ZW5N	Easy to clean, look good, love these	rating=5.0, verified=False	Before silicon counter gap covers came out I had thinner Medal ones I had to paint black, and they didn’t clean well! And Those made noise, and the paint would rub off, and I had to paint them...

Semantic results for: 'reliable stove for frequent home cooking that is easy to clean'

Rank	Score	Source	ASIN	Title	Info	Snippet
1	0.926	review	B07935N99S	Great item for easy cleanup.	rating=5.0, verified=True	These are great, especially when cooking items that overflow the pot , like soup. When a spill occurs, you simply put the burner in the sink to soak. Minimal mess left on stove. After it soaks and...
2	0.935	review	B01MT0UL8N	Exactly what I was looking for	rating=5.0, verified=True	I am a messy cook. Keeps food from falling in crack between counter and stove. Wipes clean . Stays put
3	0.965	review	B09TZQ7CLG	Good	rating=3.0, verified=False	They are a good replacement for the stove.
4	0.971	review	B0745N964K	Excellent Stove!!!	rating=5.0, verified=True	You can’t beat the price! Awesome stove and looks great!!!
5	0.974	metadata	B09KH32SSR	Singring Stove Cover Reusable Gas Stove 0.4mm thick Burner Covers Non Stick Stove Protector Easy Cleaning Liners for Gas Ranges	Tools & Home Improvement | Appliances > Parts & Accessories > Range Parts & Accessories > Accessories	Singring Stove Cover Reusable Gas Stove 0.4mm thick Burner Covers Non Stick Stove Protector Easy Cleaning Liners for Gas Ranges